Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/08/2017
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Project Title:	Eutectic Solidification in a Ternary Alloy		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35294-4461	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009-10 Materials Science Rapid Turn Around Flight Opportunity
Start Date:	08/15/2014	**End Date:**	08/14/2019
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14AP82G		
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Performance Goal Text:			
	Note this is a follow-on project for Dr. Ralph Napolitano's project,"Three-Dimensional Structures and Interface Dynamics in Univariant and Invariant Eutectic Solidification in a Ternary Alloy," which ended in 2014. Solidification processing of metallic alloys can produce a wide array of growth morphologies, which can be manipulated to access new structures, materials, and functionalities. The array of accessible structures is particularly wide in ternary and other higher-order systems, but the challenge of controlling the many variables that influence growth structures is immense and requires a better understanding of the interplay between the mechanisms of pattern formation and the influence of applied forces. Focusing on multiphase microstructures in Al-Cu-Ag ternary alloys, this experimental program is aimed at understanding and quantifying the 3D relationships between the relevant structural features, including local eutectic phase topology, crystallographic orientation/misorientation, eutectic fault configuration, faulted domain structure, and grain structure, all relative to the constraints of specimen geometry and applied thermal		

Task Description:	field. The two principal objectives to be addressed are (i) to establish an analytical framework for 2D and 3D quantitative description of multiphase microstructures, and (ii) to employ the newly developed quantitative framework to investigate the dynamics of pattern formation in ternary eutectic alloys. The work will focus primarily on the invariant three-phase eutectic structure. Ground-based samples will be processed via a vertical Bridgman-style directional solidification furnace. Electron backscatter diffraction will be a major characterization tool to measure crystallographic orientation, texture, and misorientation relationships. Analysis over the length of the sample will provide insight into initial transient behavior, grain competition, and steady state growth. The relationship between geometric and crystallographic parameters, such as curvatures, orientations, and periodic ordering, will be used to quantify the morphological trajectory of the eutectic and identify fundamental principles of pattern formation. Seeded directional solidification (DS) experiments will be conducted to compare the dynamics associated with specific crystal orientations. In order to validate the experimental correlations and analytical models developed from the findings of the ground-based work, carefully selected flight-based directional solidification experiments will be conducted using the European Space Agency (ESA) Materials Science Lab (MSL) and the Low Gradient Furnace (LGF). The project will continue to provide collaborative support to the European Space Agency's project titled "Solidification along a Eutectic Path in Ternary Alloys" (SETA), which has already accomplished important work using the Al-Ag-Cu alloy system. Significant findings of the SETA project to date have been in the areas of pattern development and the role of fault structure for both univariant and invariant eutectic compositions, the role of crystal orientation and "locked" grains, and development of nonplanar growth modes. Facilitating open scientific interaction between the Principal Investigator (PI) and the SETA team members (and between NASA and ESA), the project will be culminated with a set of flight-based microgravity experiments intended to validate the experimental correlations, mechanistic descriptions, and analytical models that are produced during the ground-based portion of this project and several other areas of investigation included within the SETA project. The planning and performance of flight-based solidification experiments is being done collaboratively, as will the analysis of post-solidification specimens upon their return to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As mentioned in the Task Description, the objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled multiphase structures through advanced solidification processing. The potential Earth-based impact of this research is far-reaching, given the extremely broad range of multiphase structures that can be fabricated through solidification processing, and is certainly not limited to metals and alloys. Indeed, the properties of nearly any multiphase material can be improved by better control of multiphase structure. In addition, the fundamental and practical knowledge from this research may lead to innovative process technologies and new materials and/or structures.
Task Progress:	Specific areas of focus during Year 3 have been (i) development and testing of code to perform automated image analysis of the three-phase eutectic structures, (ii) response of the Al-Ag-Cu three-phase invariant eutectic structure to a change in velocity, (iii) quantification of microstructural evolution in the solid state for develop of a more accurate phase field model, and (iv) continued analysis of the structures via principal component analysis. Image Analysis: Quantification of the three-phase eutectic microstructure in AlAgCu is difficult because of the variability of the structure. The standard eutectic spacing measure of lambda is not applicable because there is no obvious repeating spacing. Therefore, one activity during Year 3 was to write a series of scripts using MatLab that would measure a variety of different parameters on the eutectic images in order to discover which values correlate most strongly with processing parameters, and thereby identify the controlling factors in microstructure selection. After semi-automated thresholding, each image is automatically analyzed to determine volume fraction, area and perimeter of each phase region, specific surface area (perimeter/area), shape factor, major and minor axis length, and distances between centroids. This code has been written to calculate averages and standard deviation from either a single image or by averaging together the results from several images taken from different regions on the same cross-section in order to improve the statistical accuracy. The accuracy of the code was also checked by comparing images taken at different magnifications from the same sample and it was found that the results were consistent. In-depth analysis of existing images is ongoing. Response to Velocity Change: Continuing from year 2, a major focus during year 3 was on the behavior of ternary eutectic patterns in response to a change in solidification velocity. By perturbing an established pattern in this way, we are able to study how the system reacts to forced changes and in doing so, better identify the mechanisms that control pattern formation in this type of complex, multi-phase material. For analysis, a sample was selected that had been directionally solidified at a velocity of 0.7 μm/s for 85 mm, then the velocity quickly increased to 4.0 μm/s for an additional 50 mm. Cross sections were analyzed every 5-10 mm over that range of distances. These results were compared to another sample obtained by SETA partner DLR (German space agency), which had a velocity decrease of the same magnitude. This sample was imaged via synchrotron tomography, which permitted very detailed analysis of that sample in the region of the velocity jump, but considered only a very small total region of the sample. Finally, both of these samples were compared to 3D phase field models of the same system performed by the group of Britta Nestler at Karlsruhe Institute of Technology (KIT) in Germany. From the combined studies, a significant influence on the arising microstructure is found, leading to four main conclusions. (i) As a response to a velocity variation, a change of the phase fractions is detected in simulations as well as, at least on a short length scale, in experiments. (ii) All investigated statistics effect the velocity changes with different rates. The phase fractions adjust immediately after the velocity changes, while the reaction of the microstructural pattern occurs more slowly, which is effected by the shape factors and the nearest neighbor statistics. (iii) Despite the phase fraction variations, the overall chain-like structure remains stable. Furthermore, a tendency for an alignment in a chain-like structure is visible in the simulations. The results of both, simulations as well as experiments, indicate that there may exist stability ranges for different patterns, depending on the growth velocity. (iv) Despite the stability of the overall pattern, a continuous complex evolution of the individual rods occurs, resulting in splitting, merging, and overgrowing events as well as nucleation in the experiments. It is a particularly significant result that the analysis shows that these structures continue to evolve as they freeze,

	moving very slowly toward a more well aligned structure, but very slowly. This emphasizes the importance of long duration experiments such as long on the International Space Station (ISS). The other particularly surprising finding was the change in volume fraction after a velocity change. This effect appeared in both the experimental and the simulation results, but a conclusive reason has not been identified. The controlled environment of microgravity will be important to understand this effect better. Solid State Evolution: Over the last years, the phase field modeling group at KIT has become an important partner in this work on ternary eutectic systems, co-authoring numerous papers with the NASA PI and other SETA collaborators. One drawback to their model is that it has not incorporated solid state diffusion. It is well known that significant changes take place in this system due to the abrupt decrease in Ag solubility in the Al phase below the eutectic temperature. The only way to avoid this effect during directional solidification is to quench the sample and examine the microstructure immediate behind the quenched interface. Therefore, most experimental images of AlAgCu ternary eutectic microstructure are a combination of solidification structure and solid state diffusion effects. In order to more realistically model the physics of the real system, they are now incorporating solid state diffusion into their model. In order to directly compare the accuracy of the model to the AlAgCu system, specific experimental samples are being created by incorporating a quenched interface and evolution of the microstructure in an isothermal environment. This work is currently in progress. Principle Component Analysis: A collaboration was begun in Year 1 with Surya Kalidindi (Georgia Tech--GT) to apply their technique for principal component analysis (PCA) via pair correlations to Al-Ag-Cu three-phase eutectic microstructures. The PCA technique is used to show variation in datasets. It automatically emphasizes the factor that contributes most to the variation by plotting it along PC axis 1, the second biggest factor along PC 2, etc. Thus it requires no a priori assumptions about what feature of a dataset (such as a collection of micrographs) is changing. The GT group has published some results with other SETA partners on applying this technique to AlAgCu datasets. Although it is able to discriminate between different data sets, it is very difficult to interpret what these differences mean. In order to relate changes in PC space back to more standard stereographic measures, we are undertaking an analysis via PCA of sets of idealized microstructures based on the five theoretical patterns than were identified as possible in a directionally solidified three phase eutectic. A large set of these images was created by varying different spacings and phase fractions in a systematic way. This analysis will allow us to identify more specifically what factors shift points along PC1, PC2, or PC3, then identify the most important parameters that change with experimental condition. This analysis is ongoing.
Bibliography Type:	Description: (Last Updated: 06/05/2023)
Articles in Peer-reviewed Journals	Hötzer J, Steinmetz P, Dennstedt A, Genau A, Kellner M, Sargin I, Nestler B. "Influence of growth velocity variations on the pattern formation during the directional solidification of ternary eutectic Al-Ag-Cu." Acta Materialia. 2017 Sep 1;136:335-46. https://doi.org/10.1016/j.actamat.2017.07.007 , Sep-2017